

Your Child's Coding Journey

Coding is a long-term skill — like reading or music. The earlier a child starts, the more naturally it becomes part of how they think and solve problems. This guide shows you what that journey looks like at Spark Coding, from a child's first lines of code through to the kinds of opportunities open to teenagers who keep at it.

AT A GLANCE

- A clear, age-mapped path from Year 3 through Year 12.
- Personalised learning, with a maximum of 3 students per class.
- Optional doors along the way — competitions, scholarships, real-project portfolios — none of which need to be decided up front.
- Honest about timing: most children spend one to two years at each stage.

The stages of the journey

Four stages, each building on the last. Most children spend one to two years at each. Older beginners often skip the earlier stages entirely.

Stage 1 — First Code

AGES 8–9 · YEAR 3

Learning to think in code through Scratch, building animations and simple games. Children move from *playing* games to *making* them — a quietly transformative shift in how they see technology.

By the end: can build a complete game with score-keeping, multiple levels, and characters that respond to the player.

Stage 2 — Bridge to Real Code

AGES 9–10 · YEAR 3 TO YEAR 4

Starting to read actual programming text alongside the friendly coloured blocks. The first taste of what professional programmers see on screen — without losing the safety net of blocks they already know.

By the end: can recognise the same code in two forms — blocks and Python — and make small text edits with confidence.

Stage 3 — Python

AGES 10–12 · YEAR 4 TO YEAR 6

The same language used in real software, AI, and at university. Children write Python from scratch and build small, useful programs — quizzes, calculators, data tools — they can show off to family and friends.

By the end: a portfolio of half a dozen working programs they wrote themselves, and the vocabulary to read code in any modern language.

Stage 4 — Building Skills

AGES 12+ · YEAR 7 ONWARDS

Apps, websites, and more sophisticated games. For students who want it, preparation for national programming competitions — and the long path that opens up from there.

By the end: a substantial public portfolio of real projects; the choice of whether to take coding further is fully in the student's hands.

A note for older beginners

If your child is starting in Year 5 or later, we don't rewind them through the earlier stages. A Year 7 beginner aiming at career preparation typically begins in Stage 3 (Python) directly, building real programs from the very first lesson. The stages above are a map, not a queue.

How we choose where to start

The first 45-minute trial session doubles as a placement check. We see what your child already knows, what excites them, and what kind of project would feel meaningful to build. By the end of the trial, we'll have a clear, personalised plan for the next term.

Where this can lead

Three doors that open progressively. None are required — your child can step through any, all, or none, and still come out a stronger thinker than they started.

DOOR 1 · AVAILABLE TO EVERY STUDENT

A stronger learner

Computational thinking is now part of the NSW curriculum. Students who code think more clearly, debug their own work, and break big problems into smaller ones — skills that show up in maths, science, and even essay writing. This door opens automatically; your child doesn't need to win anything to walk through it.

DOOR 2 · FROM AROUND YEAR 3, OPT-IN

Friendly national competitions

Australia runs short online competitions every year — fun, low-cost, and designed to be enjoyable. The well-known ones: **Bebras** (Feb), **CAT** (May), **AMC** (Aug), and from Year 7, the **Australian Informatics Olympiad**. We register and prepare students in class — no extra work at home.

DOOR 3 · FROM YEAR 7 ONWARDS, REQUIRES SUSTAINED EFFORT

Scholarships, university, and beyond

For students who keep coding seriously into high school, sustained competition results are recognised by private-school scholarship panels, Australian university bonus-point schemes, and international university applications for those who reach the top levels. Just as importantly, the **portfolio of real projects** students build with us becomes a standalone signal — recognised from Year 9 work-experience applications through to university interviews.

How you'll know your child is on track

AFTER ONE TERM

They'll show you a small program they built themselves. You'll start hearing them ask: *"What if I made it do this instead?"*

AFTER ONE YEAR

They'll be fixing their own bugs without needing to ask. They'll have made something they're genuinely proud of, and they'll want to keep going.

AFTER THREE YEARS

They'll be reading and writing the same kind of code professional programmers use. They'll have at least one competition certificate. They'll know — from their own experience — whether coding is "a fun side thing" or "something they want to keep doing."

What the journey actually looks like

Three real-shaped examples. Each child started at a different age and took a different path — all of them ended up confident and capable.

Mia — started Year 3

First year: built her first Scratch animations and sat Bebras for fun. Year 4: made her first arcade-style game and joined the school's CAT cohort. Year 5: moved into Python and built a recipe-recommendation app she showed at the school showcase. By Year 7, she's writing real programs and sitting her first Australian Informatics Olympiad. By Year 12, she has a six-year portfolio of projects and competition results that strengthens any university application — whether she chooses STEM or not.

Eli — started Year 6

Came in with no coding experience but wanted to "get a head start" before high school. Spent his first term in Python building real, small programs (a quiz game, a Mad Libs generator, a personal website). By the end of Year 7 he sat his first Australian Informatics Olympiad. By Year 9, he's comfortable enough to mentor younger students. He treats coding as a long-term hobby that keeps options open — university STEM degrees become a comfortable choice, not a forced one.

Aanya — started Year 4

Did Scratch and the bridge-to-Python stage through Years 4 and 5, picked up Python in Year 6, with AMC and Bebras every year. Sat for a private-school scholarship at the end of Year 6 — the panel specifically asked about her Bebras and AMC results, and the GitHub portfolio of her Python projects came up in the interview. Whatever the scholarship outcome, she enters Year 7 confident and capable.

WHERE YOUR CHILD IS NOW

Current stage

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Next milestone

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What we'll work on next term

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Ready to start the conversation?

We offer a free 45-minute trial session — no commitment required. Onsite at our Rouse Hill studio, or online via Google Meet. Ongoing classes are kept to a maximum of three students per group at \$59 per session, billed per term.

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